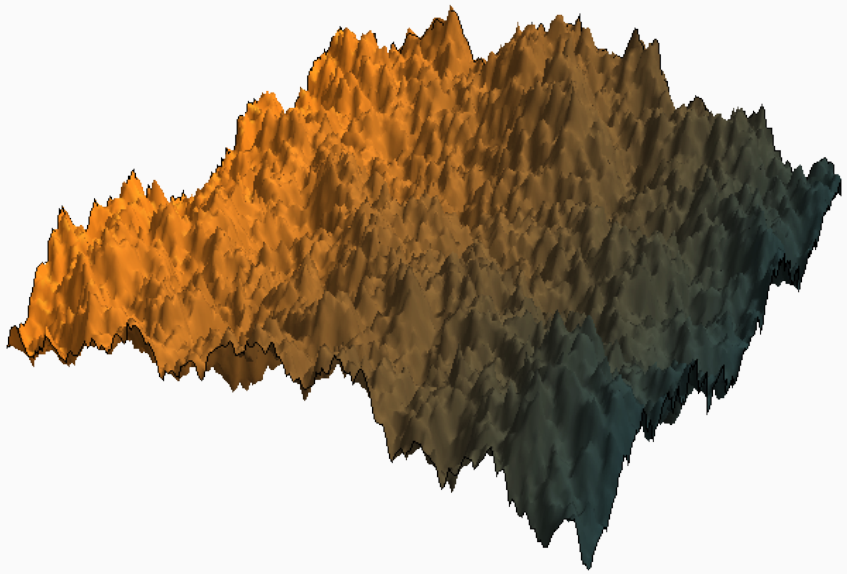
Quantum Field Theories
as
Statistical Field Theories



Antoine Tilloy
Max Planck Institute of Quantum Optics, Germany

Theory seminar, **Wigner Research Center,** Budapest, Hungary
February 17, 2017

MPQ

Why is there a problem?

"We know that the moon is demonstrably not there when nobody looks"



David Mermin 1981

The only connexion between the **formalism** of quantum theory and **Nature** is through the measurement postulate.

"A **mathematically trivial** operation"

Measurement postulate

For a system "described" by $|\psi\rangle \in \mathscr{H}$ and a measurement of orthogonal projectors Π_i s. t. $\sum_i \Pi_i = \mathbb{1}$ one has:

Born rule :

Result " i " with probability $\mathbb{P}[i] = \langle\psi| \Pi_i |\psi\rangle$

Collapse :

$$|\psi\rangle \longrightarrow \frac{\Pi_i |\psi\rangle}{\sqrt{\mathbb{P}[i]}}$$



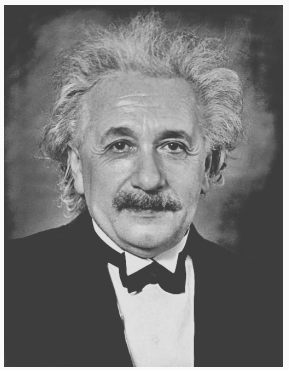
Max Born 1926



John von Neumann 1932

“A **physically subtle** endeavour”

- What is a measurement?
- How can measurement be a primitive concept?
- What is a measurement result made of?
- Can one deduce the measurement postulate from unitary evolution?



Albert Einstein 1935



John S. Bell 1989

notions of ‘reversible’ and ‘irreversible’. Einstein said that it is theory which decides what is ‘observable’. I think he was right – ‘observation’ is a complicated and theory-laden business. Then that notion should not appear in the *formulation* of fundamental theory. *Information? Whose* information? Information about *what*?

On this list of bad words from good books, the worst of all is ‘measurement’. It must have a section to itself.

Physics World, Against Measurement

The curse of linearity in a “measurement” situation

Initial state:

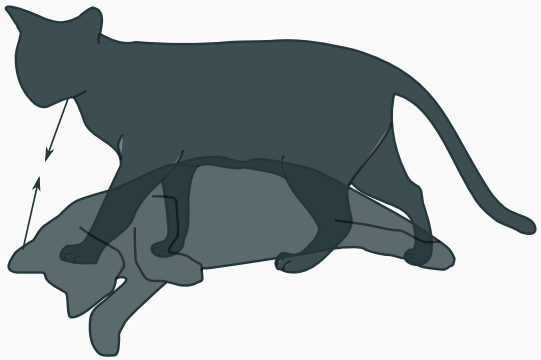
$$|\Psi\rangle_0 = \frac{1}{\sqrt{2}}\Big(|\uparrow\rangle + |\downarrow\rangle \Big) \otimes |\spadesuit\rangle \quad \in \mathbb{C}^2 \otimes \mathbb{C}^{2^{10^{23}}}$$

After evolution:

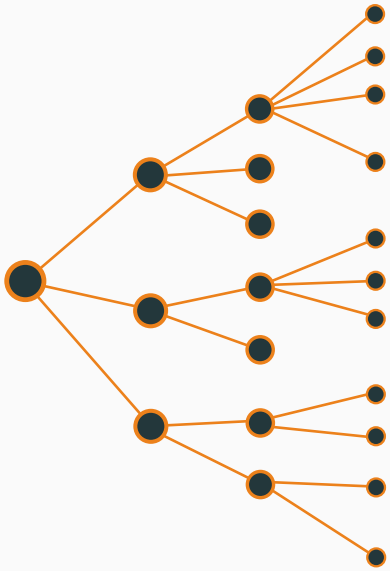
$$|\Psi\rangle_t = \frac{1}{\sqrt{2}} |\uparrow\rangle \otimes |\diamondsuit\rangle + \frac{1}{\sqrt{2}} |\downarrow\rangle \otimes |\heartsuit\rangle$$

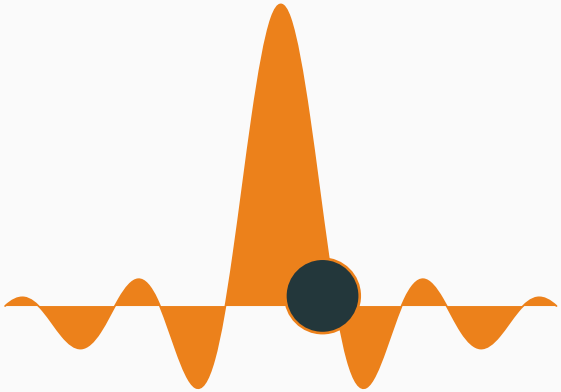
$\rightarrow$ typically $|\clubsuit\rangle \perp |\heartsuit\rangle$: **decoherence**

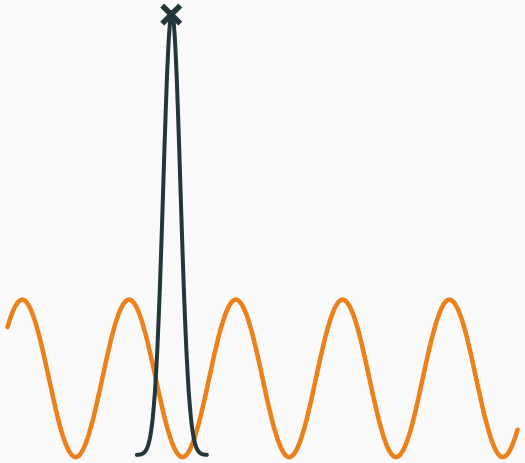
$\rightarrow$ but **no collapse** because of linearity



Reformulations of quantum theory are needed









Everett 1957



Bohm 1952



Ghirardi 1986

In Quantum field theory

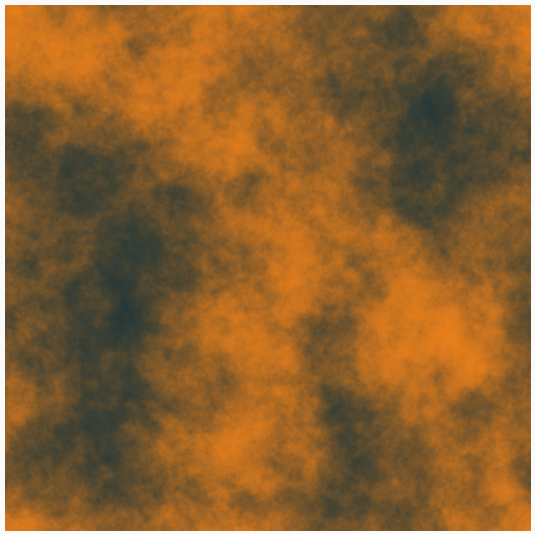
First problem:

Quantum **field** theory is not about **fields**, there are no fields, it's about correlation functions of … macroscopic stuff made of …

In Quantum field theory

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Quantum **field** theory is not about **fields**, there are no fields, it's about correlation functions of ... macroscopic stuff made of ...



what we want

$$\langle \hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\rangle = \sum \cdots$$

$$\langle \hat{\psi}(x_1)\hat{\psi}(x_2)\hat{\psi}(x_3)\hat{\psi}(x_4)\rangle = \sum \cdots$$

what we get

In Quantum field theory

Second problem:

Even as operational theories, “physical” interacting quantum field theories are ill defined.

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Even as operational theories, “physical” interacting quantum field theories are ill defined.

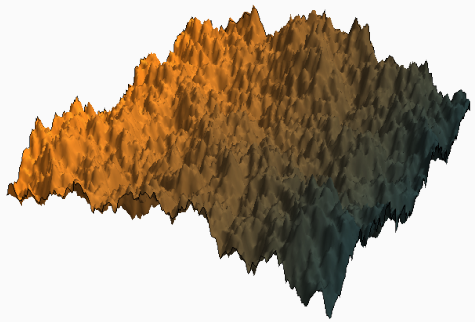
$$\langle \hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\rangle = \sum_{k=0}^{+\infty} (\infty)_k \, g^k$$

$$\langle \hat{\psi}(x_1)\hat{\psi}(x_2)\hat{\psi}(x_3)\hat{\psi}(x_4)\rangle = \sum_{k=0}^{+\infty} (\infty)_k \, g^k$$

These are actually the two classes of difficulties noted by Paul Dirac in 1963

QUESTION

Can we reformulate QFT as a theory of “true” fields?



Advantages:

- naturally solves the conceptual difficulty of quantum theory
- may provide a way to understand divergences

Idea in a nutshell

1. For a quantum scalar field:

$$\left[\hat{\phi}(x), \hat{\phi}(y)\right] \neq 0$$

So $\phi(x)$, the "measured" value of $\hat{\phi}(x)$, cannot be defined everywhere in $\mathbb{R}^4$. $\Rightarrow$ a quantum field is not "objective" stuff.

2. However, one can construct ξ a complex classical scalar field such that:

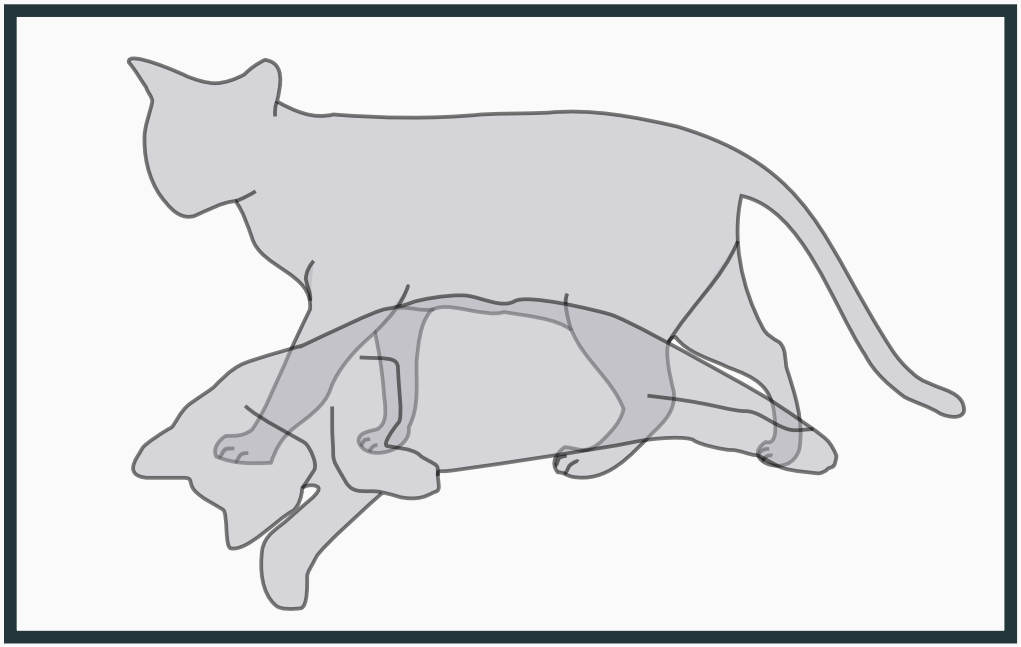
$$\underbrace{\mathbb{E}[\xi(x)\xi^*(y)]}_{\text{statistical average}} = \underbrace{\langle\hat{\phi}(x)\hat{\phi}(y)\rangle}_{\text{quantum expectation}}$$

3. Does not give all correlation functions **but** typically encodes **all the dynamics** on other quantum degrees of freedom for linearly coupled quantum systems.

A RECIPE FOR COLLAPSE MODELS

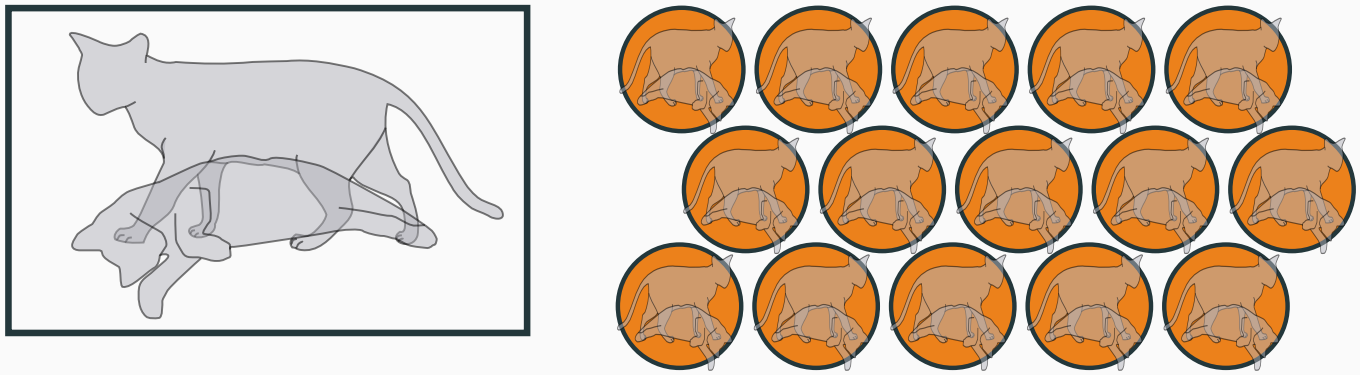
Start with a quantum system

$$|\psi(t)\rangle$$



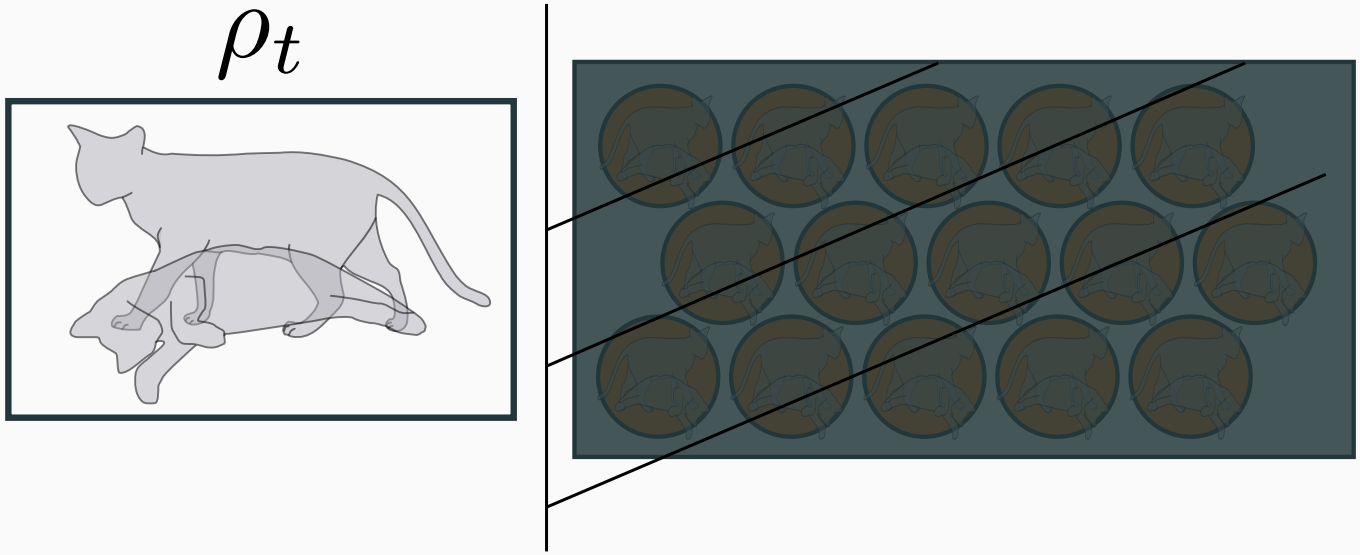
Couple it with a bosonic bath

$$|\psi(t)\rangle \otimes |\phi(t)\rangle$$



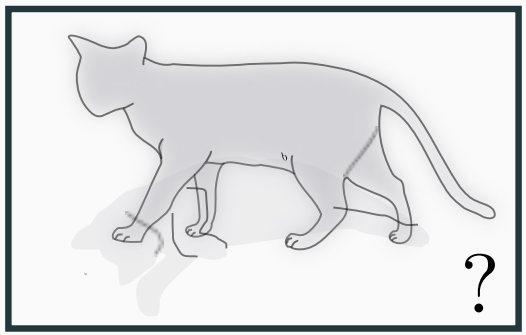
A RECIPE FOR COLLAPSE MODELS

Trace out the bath

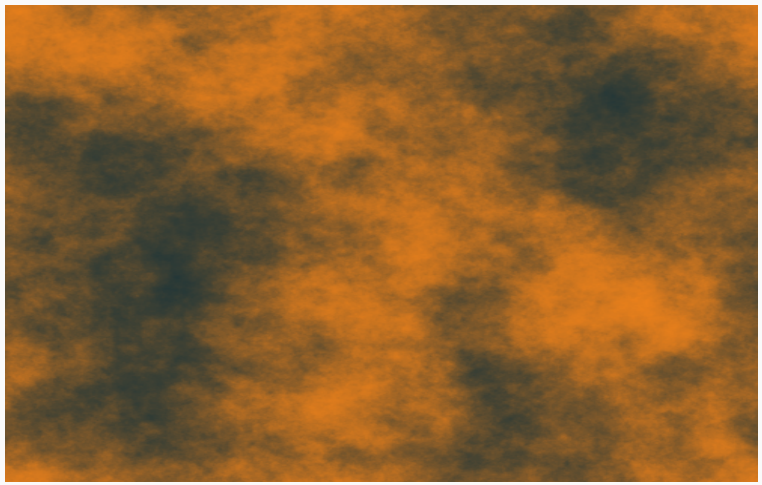


A RECIPE FOR COLLAPSE MODELS

$|\psi_w(t)\rangle$

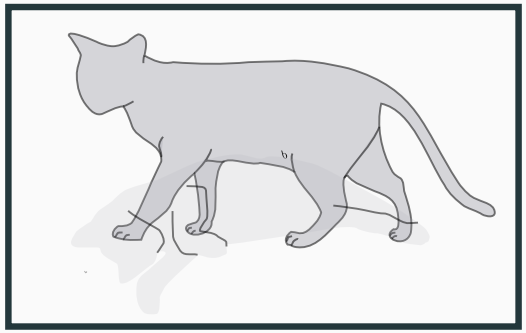


$\mathrm{d}\mu_0(w)$

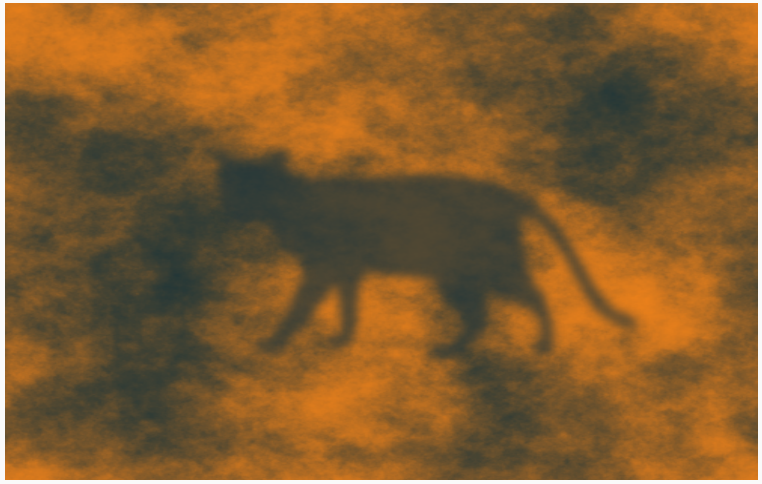


Normalize the state and redefine the probability measure

$$|\tilde{\psi}_w(t)\rangle$$



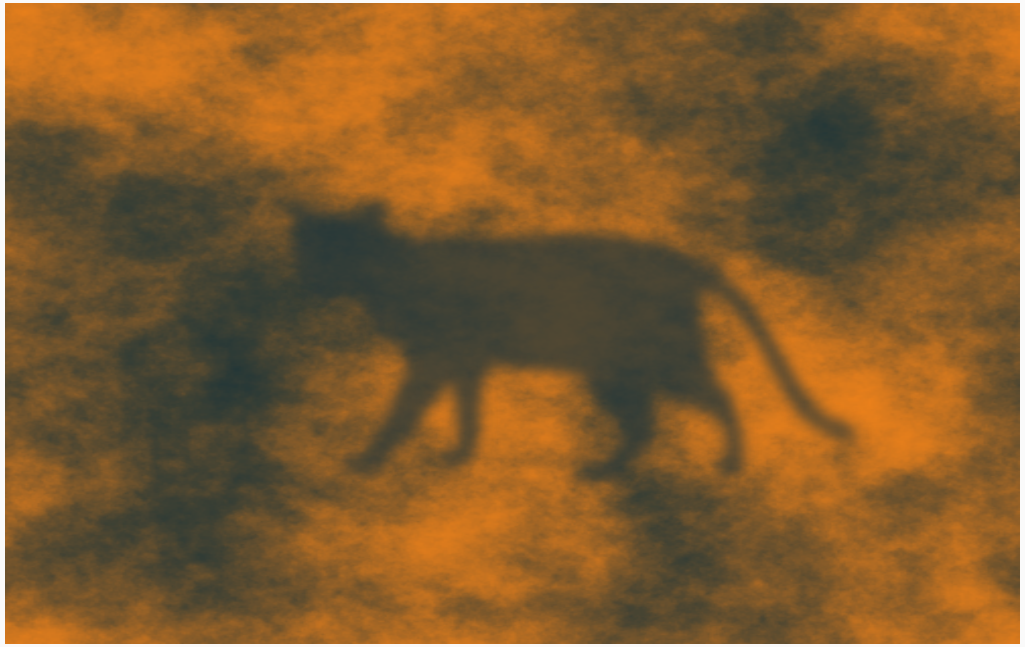
$$\mathrm{d}\mu_t(w)$$



A RECIPE FOR COLLAPSE MODELS

Keep only the stochastic field

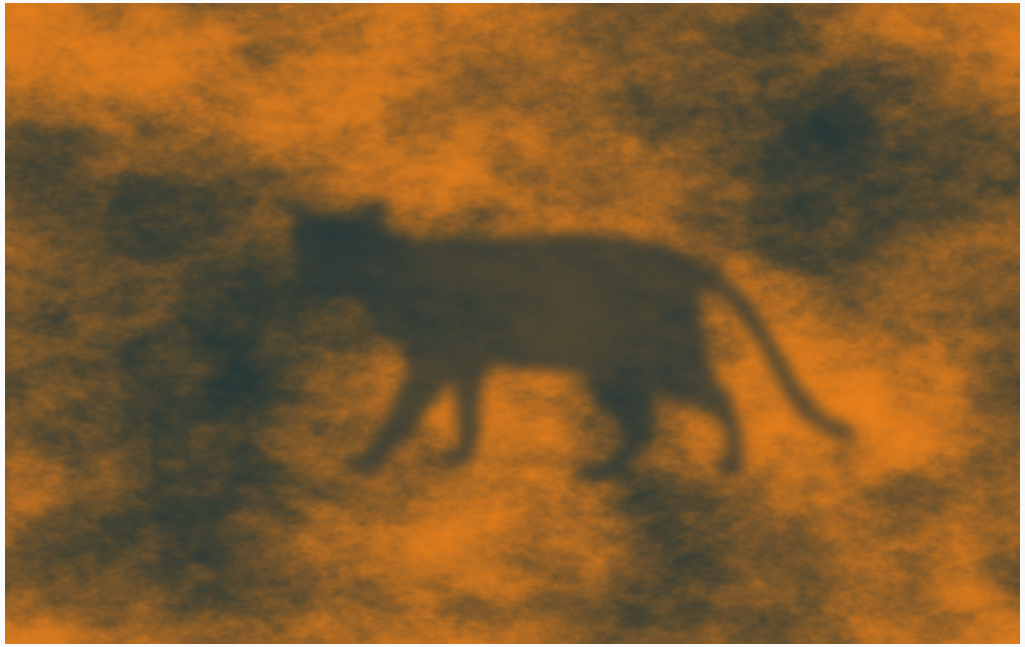
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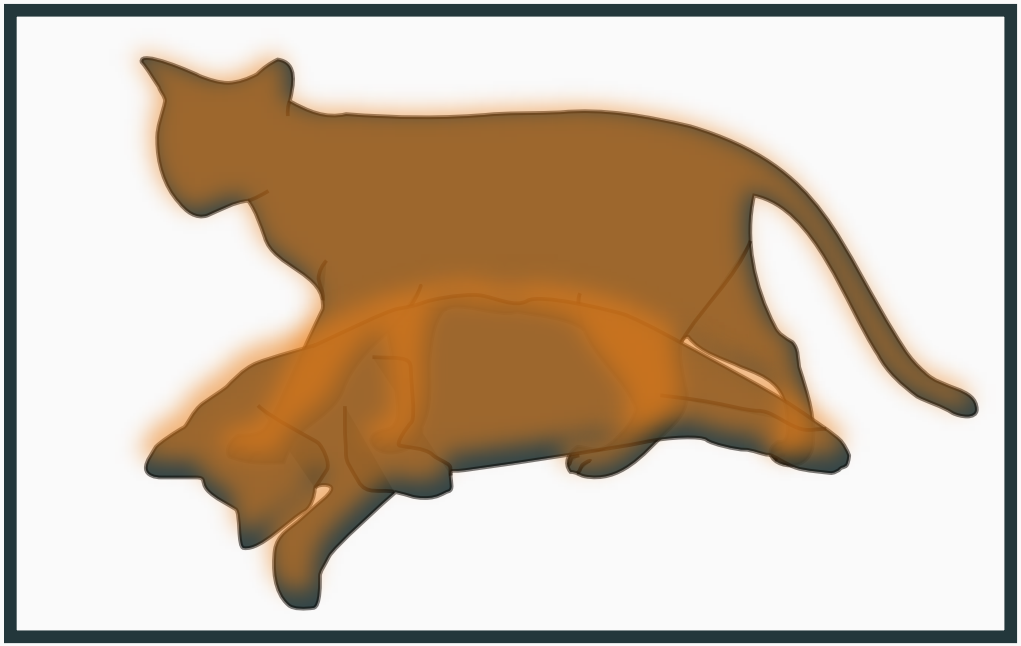


$\rightarrow$ you now have a single cat, living in $\mathbb{R}^3$

CHANGING THE RECIPE FOR QFT

Take an interacting QFT:

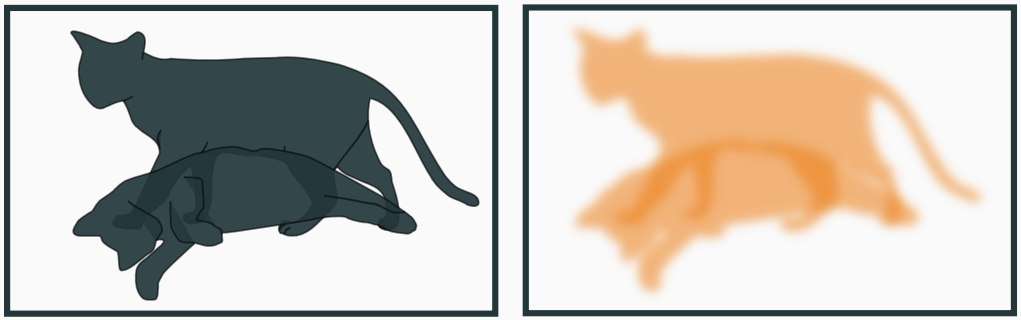
$$|\Psi(t)\rangle$$



CHANGING THE RECIPE FOR QFT

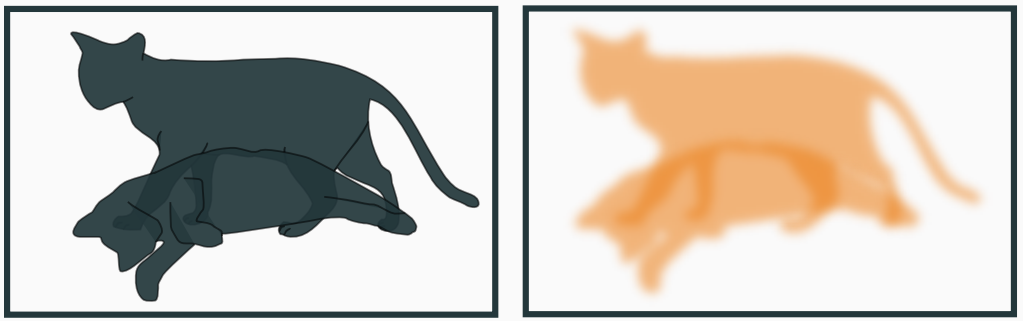
Split fermions and bosons:

$$|\psi(t)\rangle \quad \otimes \quad |\phi(t)\rangle$$



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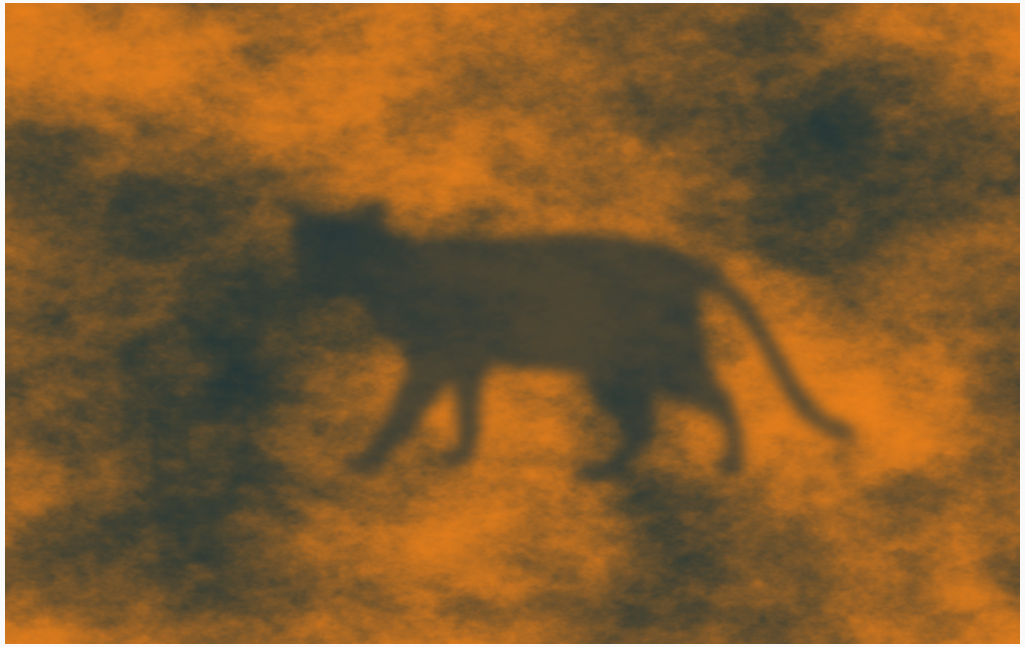
$$|\psi(t)\rangle \quad \otimes \quad |\phi(t)\rangle$$



$\rightarrow$ now do the same as before with the bosons as bath

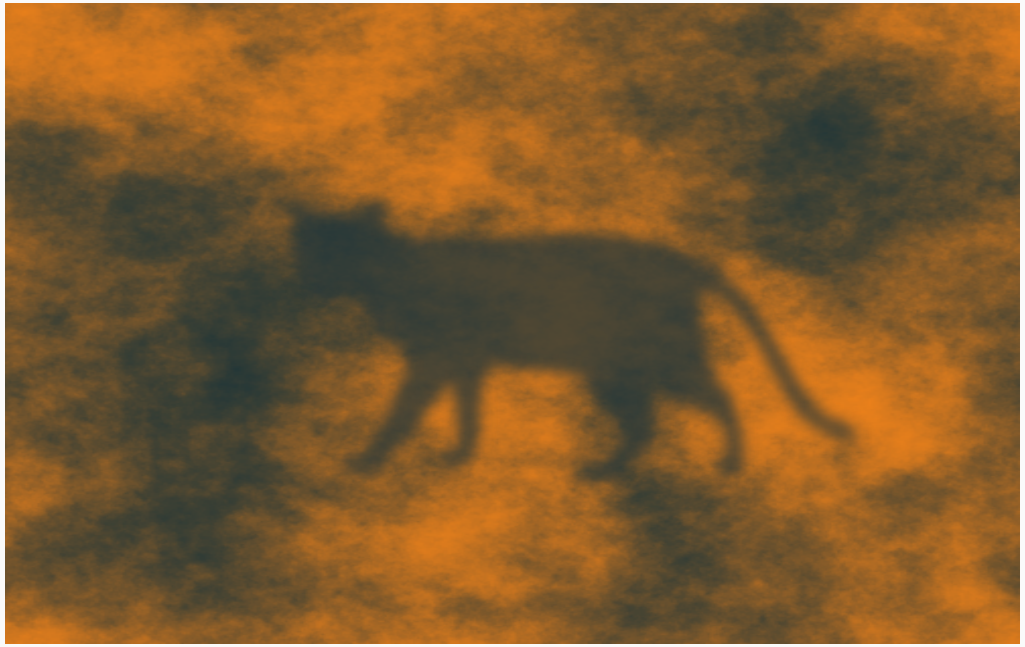
Final picture:

$$\mathrm{d}\mu_t(w)$$



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$$\mathrm{d}\mu_t(w)$$



This time without changing the empirical content of the theory!

Consider a Yukawa theory:

$$\begin{aligned}
\mathscr{L}_f(\psi, \partial_\mu \psi) &= \bar{\psi}(i\partial\!\!\!/ - m_f)\psi \\
\mathscr{L}_{\text{int}}(\psi, \phi) &= g\,\bar{\psi}\,\psi\,\phi \\
\mathscr{L}_b(\phi, \partial_\mu \phi) &= \frac{1}{2}\partial_\mu \phi\,\partial^\mu \phi - \frac{1}{2} m_b^2 \phi^2,
\end{aligned}$$

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$$\hat{\rho}_f(t) = \mathcal{T} \exp\left(i\Phi\left[j^L, j^R\right]\right) \cdot \hat{\rho}_f(t_i)$$

where $\Phi[j^L, j^R]$ is the "operator" influence phase functional:

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$$i\Phi\left[j^L, j^R\right] = \int_{t_i}^{t}\int_{t_i}^{t} \mathrm{d}^4x\,\mathrm{d}^4y\; D(x,y)\, j^L(x) j^R(y)$$
$$- \frac{1}{2}\theta(x^0 - y^0)\, D(x,y)\, j^L(x) j^L(y) - \frac{1}{2}\theta(y^0 - x^0)\, D(x,y)\, j^R(x) j^R(y)$$

with $D(x,y) = \mathrm{tr}_b\left[\hat{\phi}(x)\hat{\phi}(y)\,\hat{\rho}_b(t_i)\right] = \langle\hat{\phi}(x)\hat{\phi}(y)\rangle$.

2) Unraveling the open evolution $\mathbb{E}\big[\,|\psi_\xi\rangle\,\langle\psi_\xi|\,\big] = \hat{\rho}_f(t)$

$$
\begin{aligned}
|\psi_\xi(t)\rangle = & \mathcal{T}\exp\Bigg\{ -i\int_{t_i}^{t} \mathrm{d}^4x\, \hat{\jmath}(x)\,\xi(x) \\
& - \int_{t_i}^{t}\int_{t_i}^{t} \mathrm{d}^4x\,\mathrm{d}^4y\,\theta(x^0-y^0)\,[D-S](x,y)\,\hat{\jmath}(x)\hat{\jmath}(y)\Bigg\}\,|\psi_\xi(t)\rangle
\end{aligned}
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with

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$$

Note: importantly, finding ξ with the correlation matrix D is always possible (the relation matrix S is a free parameter)

3) Cook the measure

$$\mathrm{d}\mu_t(\xi) = \langle \psi_\xi(t) | \psi_\xi(t) \rangle \mathrm{d}\mu_{t_i}(\xi)$$

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Finally push $t_i \to -\infty$, $t \to +\infty$

In functional integral representation, with $\Psi = (\psi, \bar{\psi})$:

$$\mathrm{d}\mu_{+\infty}(\xi) = \int \mathcal{D}[\Psi]\mathcal{D}[\Psi']\, \rho_f(\Psi_{-\infty}, \Psi'_{-\infty})\, \delta(\Psi_{+\infty}, \Psi'_{+\infty})$$
$$\kappa^*[j']\, \kappa[j] \exp\left(i\mathscr{S}_f - i\mathscr{S}'_f - i\int j\xi - j'\xi^* \right) \underbrace{\mathrm{d}\mu_{-\infty}(\xi)}_{\text{Gaussian measure}} ,$$

We get a non-Gaussian deformation of the probability measure

Recollecting the ideas

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APARTÉ WITH COLLAPSE

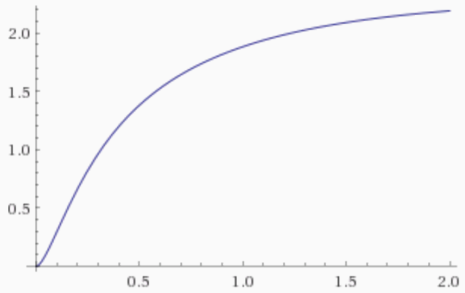
One can prove **collapse** in position and **amplification**

APARTÉ WITH COLLAPSE

One can prove **collapse** in position and **amplification**

Collapse "strength" Ω between two points x and y:

$$\Omega_{+\infty}(\mathbf{x},\mathbf{y}) = \frac{-g^2}{(2\pi)^2}\left(K_0\left(m_b|\mathbf{x}-\mathbf{y}|\right) - K_0\left(\Lambda|\mathbf{x}-\mathbf{y}|\right) - \log\left[\frac{\Lambda}{m_b}\right]\right).$$

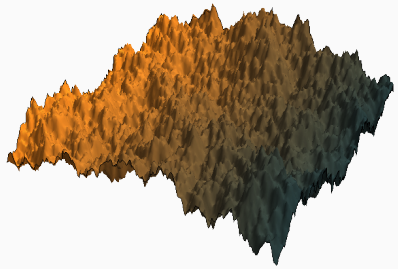


RECONSIDERING QFT

1) QFTs can be seen as theories about fields (without Wick rotation)

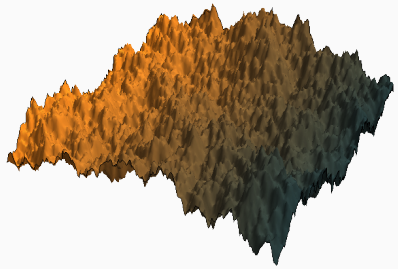
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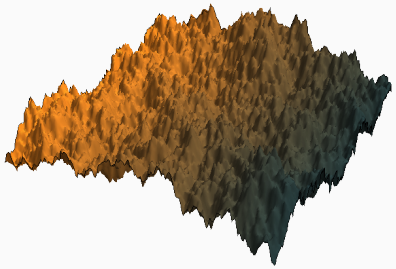
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2) QFTs can be fundamentally UV-regularized

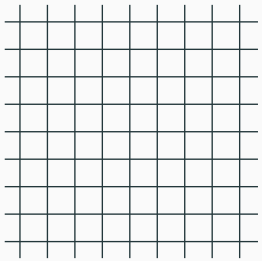
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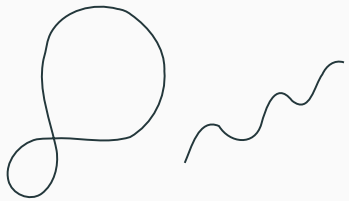
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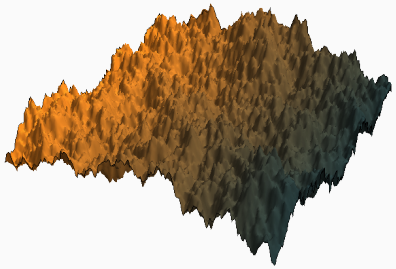
No Ostrogradsky instability, i.e. no need for:





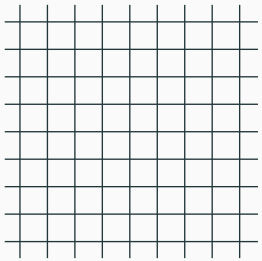
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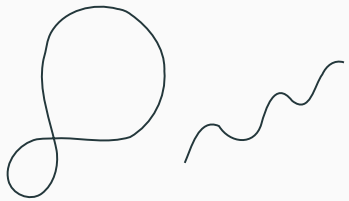
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→ makes Wilsonian renormalization transparent

LIMITATIONS

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- It would be better to work from first principles